MATH 301: Homework 5

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1 Question 1

Show that every normed vector space $(V, \|\cdot\|)$ is automatically a metric space (V, d) with the metric d given by $d(x, y) = \|x - y\|$

We demonstrate all principles of a metric; that is for all $x, y, z \in V$ we have:

Property	Description
Identity of Indiscernibles:	$d(x, y) = 0 \iff x = y$
Symmetry	$d(x, y) = d(y, x)$
Triangle Inequality	$d(x, y) \leq d(x, z) + d(z, y)$
Non-Negatitivity	$d(x, y) \geq 0$

Table 1: Your table caption here